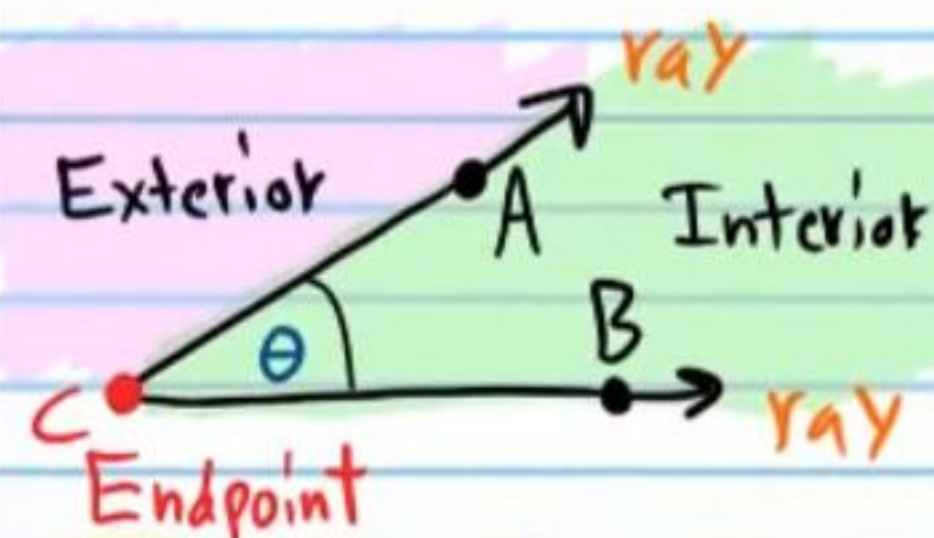


Fundamental Geometry (Part II)

Name
Date
Course

Angles

Angle: Two **rays** with a common **endpoint** that help to indicate a measure of rotation.



Three Ways to Notate an angle.

$\angle ACB$, $\angle BCA$, $\angle C$

θ is the measure of rotation.

Angle Vertex: The common **endpoint** of the two **rays**.

Angle Sides: The two **rays**.

The measure of an angle is the extent of rotation, and its most common unit is the degree (notated with a "°" superscript). 360 degrees (i.e. 360°) is defined as the measure of a complete rotation.

Four Ways to Notate the Measure of an Angle

$m\angle ACB$, $m\angle BCA$, $m\angle C$, or a Greek letter (e.g. θ)

Practice

① i) Write three names for the angle below. ii) Write four names for the measure of the angle. iii) Identify its sides and vertex.

i) $\angle EHC$, $\angle CHE$, $\angle H$

ii) $m\angle EHC$, $m\angle CHE$, $m\angle H$, θ

iii) Vertex: H Sides: $\vec{HE}$, $\vec{HC}$

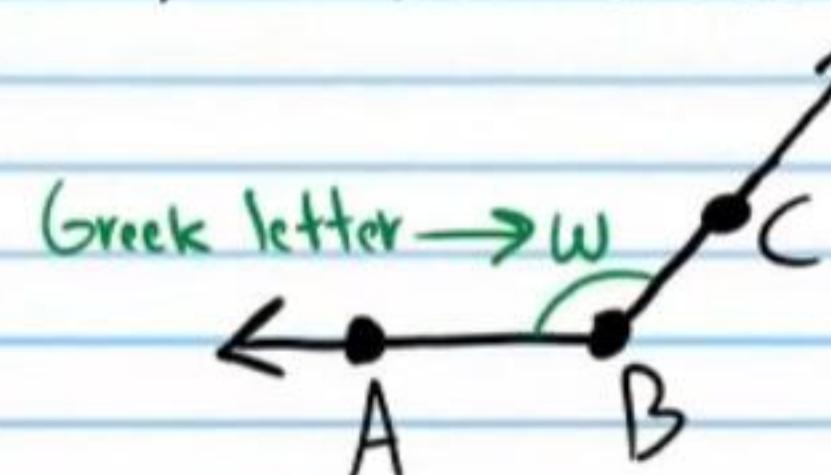


② i) Write three names for the angle below. ii) Write four names for the measure of the angle. iii) Identify its sides and vertex.

i) $\angle ABC$, $\angle CBA$, $\angle B$

ii) $m\angle ABC$, $m\angle CBA$, $m\angle B$, w

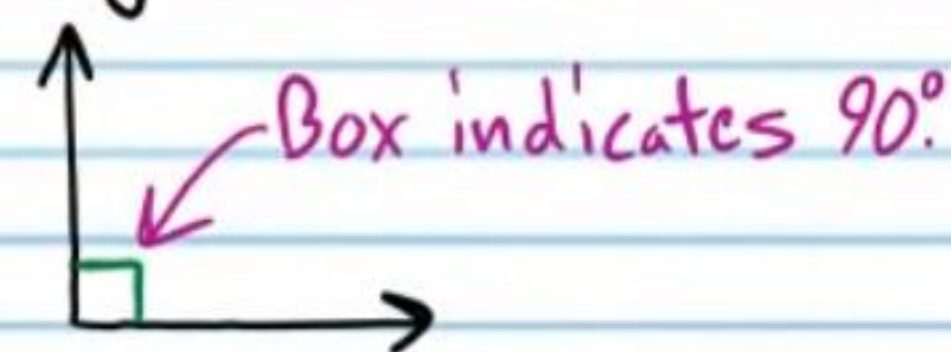
iii) B, $\vec{BA}$, $\vec{BC}$



Classifying Angles

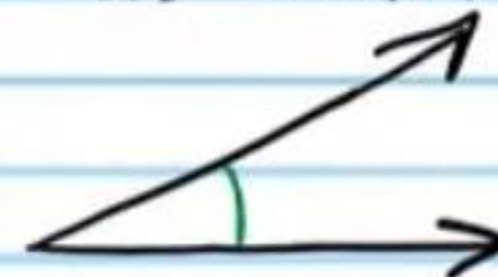
Right Angle

An angle whose measure is 90°



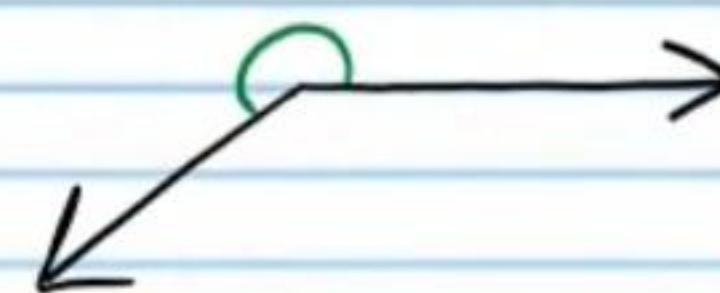
Acute Angle

An angle whose measure is less than 90°



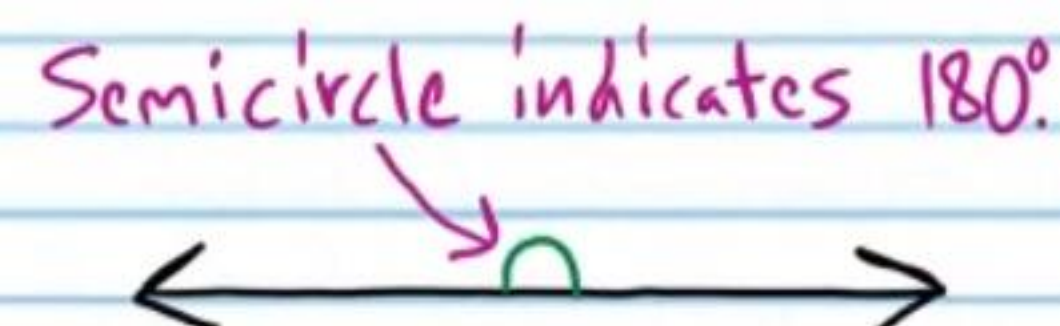
Reflex Angle

An angle whose measure is between 180° and 360°



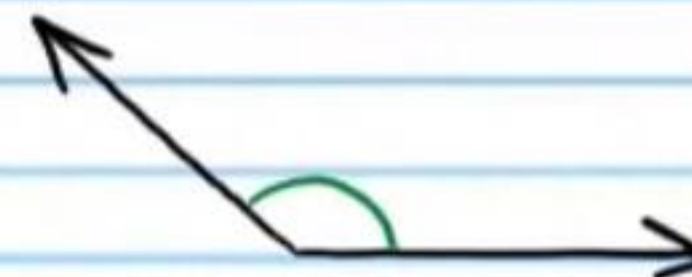
Straight Angle

An angle whose measure is 180°



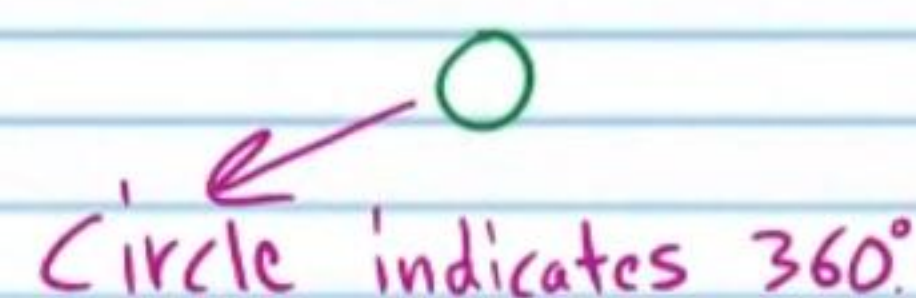
Obtuse Angle

An angle whose measure is between 90° and 180°



Complete Rotation Angle

An angle whose measure is 360°



Practice

③ Classify the angles with the following measures.

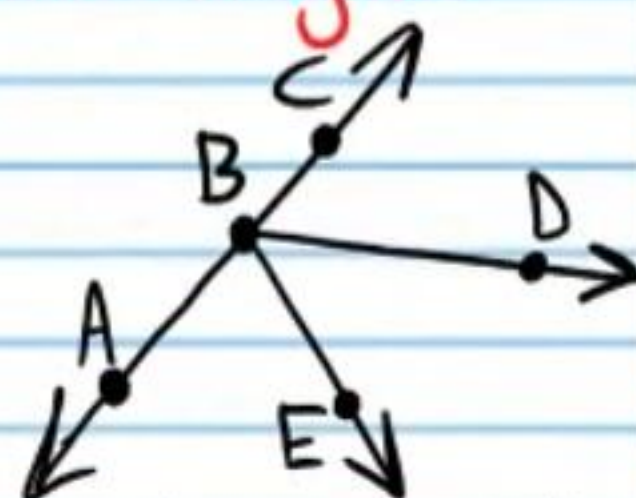
a) 190° b) 180° c) 150° d) 90° e) 89° f) 309°

Reflex Straight Obtuse Right Acute Reflex

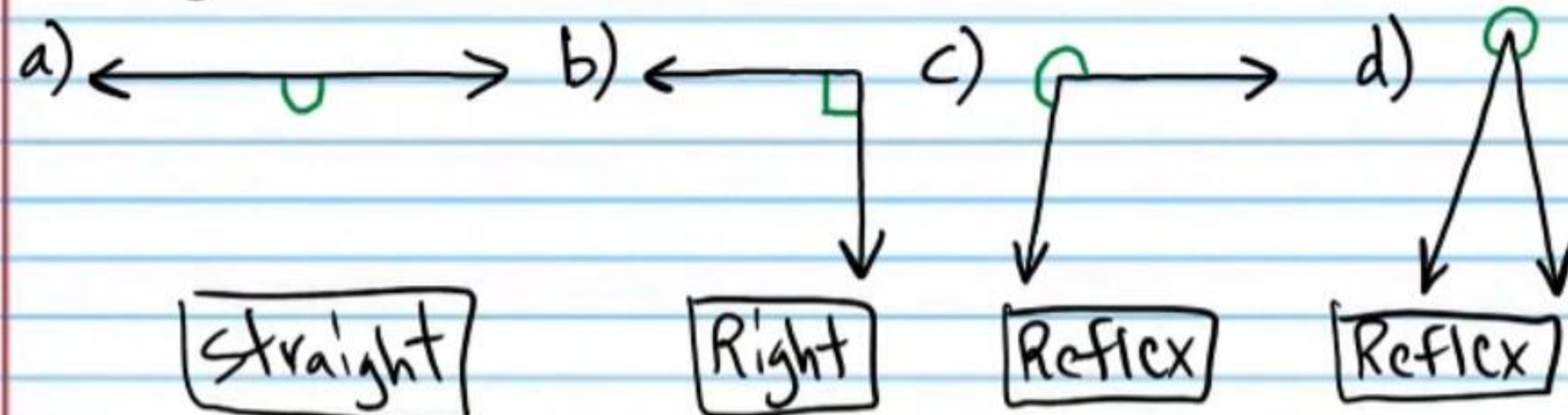
④ Use the illustration to classify each angle below.
Assume that none of these are reflex angles.

$\angle ABD$, $\angle ABC$, $\angle CBD$, $\angle CBE$

Obtuse Straight Acute Obtuse

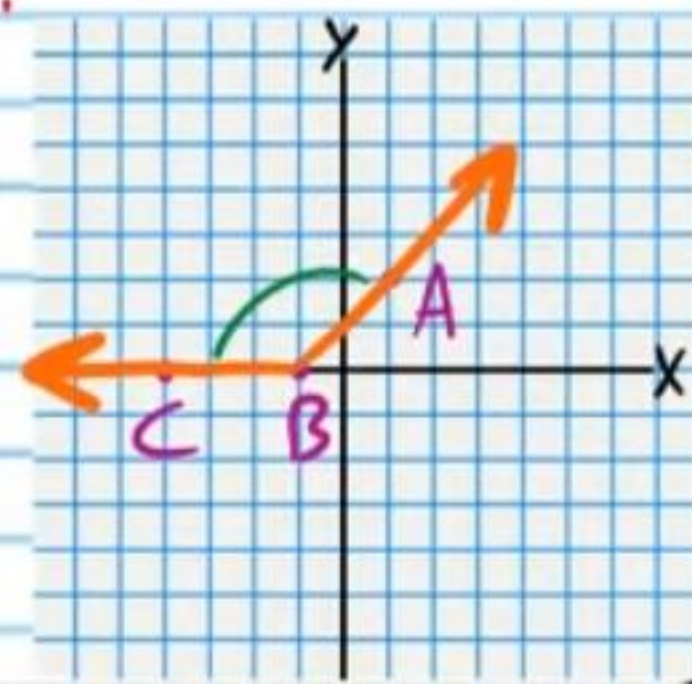


⑤ Label each angle below as acute, obtuse, right, straight, reflex, or complete rotation.



⑥ Classify $\angle ABC$ with $A(1,2)$, $B(-1,0)$, and $C(-4,0)$.
Assume that it is not a reflex angle.

Obtuse



⑦ Classify the angles with the following measures.

a) 11° b) 91° c) 200° d) 180° e) 90° f) 73°

Acute Obtuse Reflex Straight Right Acute

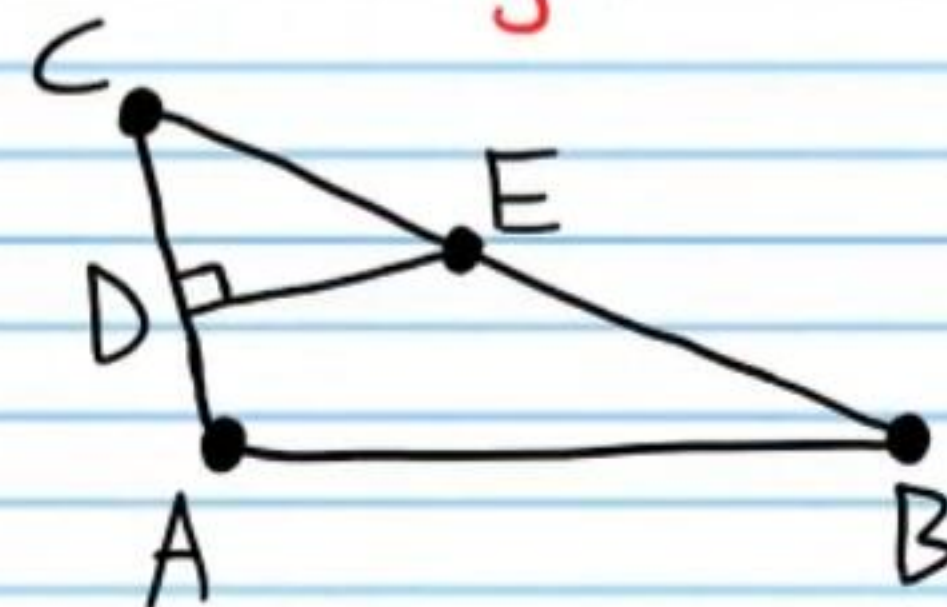
⑧ Use the diagram to classify each angle below.
Assume that none of these are reflex angles.

$\angle ABE$, $\angle CDE$, $\angle CAB$,

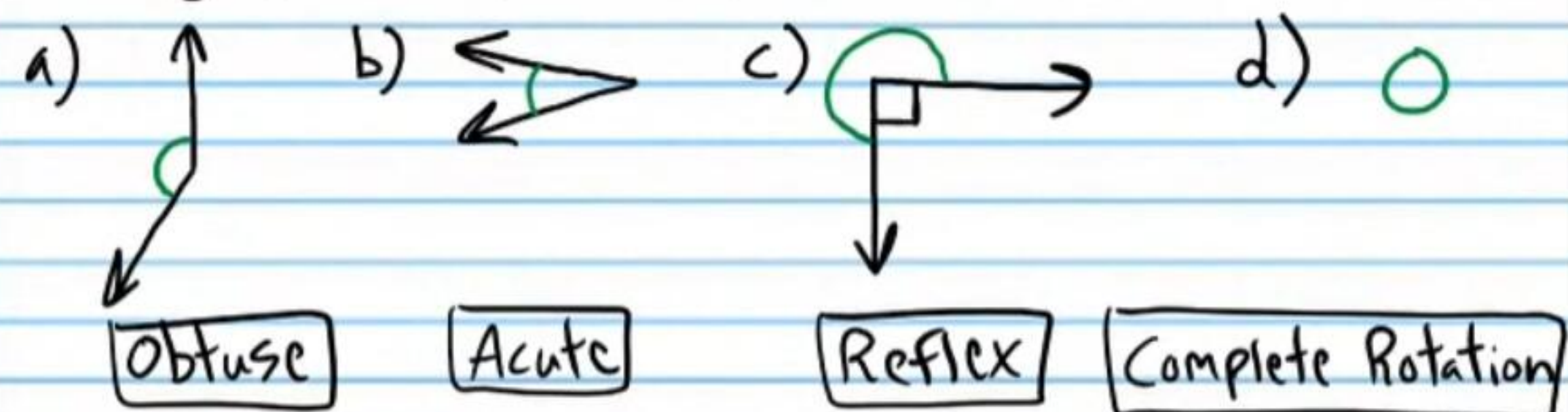
Acute Right Obtuse

$\angle DEB$, $\angle DCE$, $\angle EDA$

Obtuse Acute Right

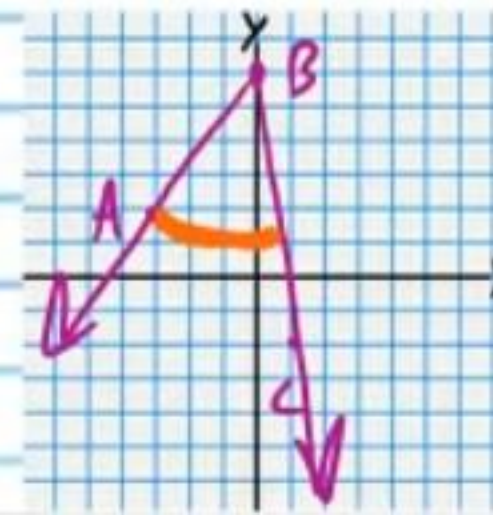


⑨ Label each angle below as acute, obtuse, right, straight, reflex, or complete rotation.



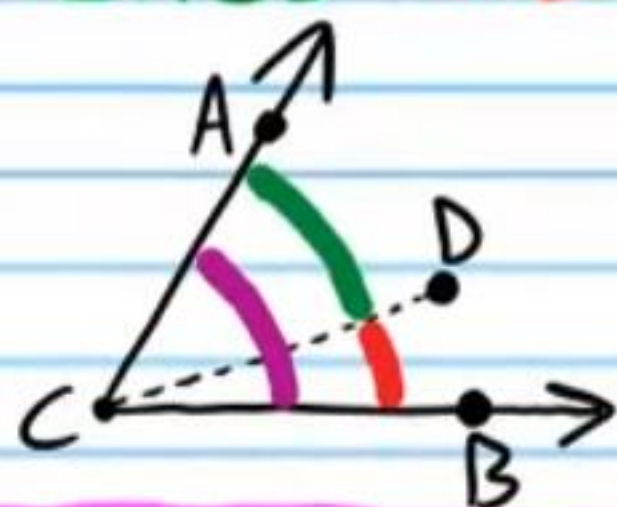
⑩ Classify $\angle ABC$ with $A(-3,2)$, $B(0,6)$, and $C(1,-2)$.
Assume that it is not a reflex angle.

Acute



Angle Addition Postulate

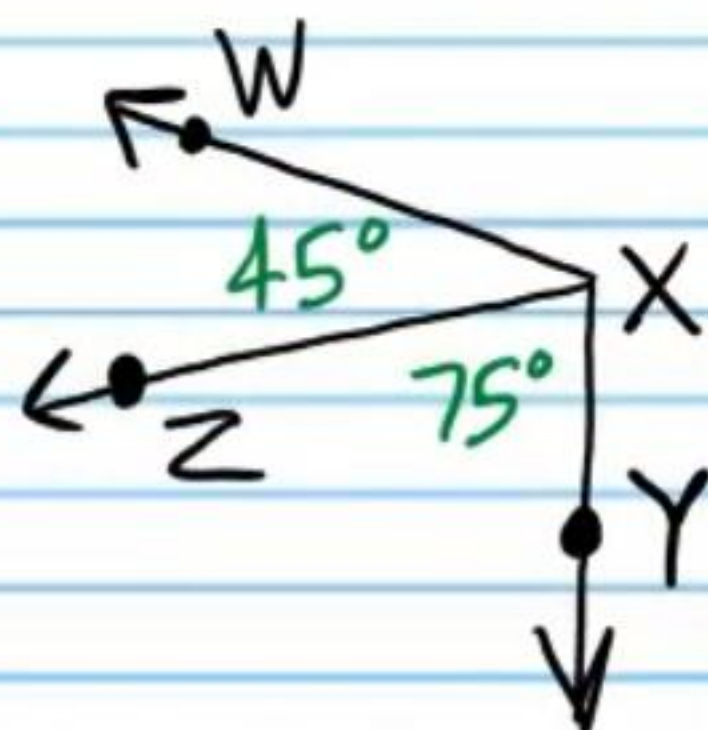
If D is in the interior of $\angle ACB$, then
 $m\angle ACB = m\angle ACD + m\angle DCB$.



Practice

⑪ Find $m\angle WXY$.

$$\begin{aligned} m\angle WXY &= m\angle WXZ + m\angle ZXY \\ &= 45^\circ + 75^\circ \\ &= \boxed{120^\circ} \end{aligned}$$



⑫ If $m\angle ABC = 47^\circ$, find $m\angle DBC$.

i) $m\angle ABC = m\angle ABD + m\angle DBC$

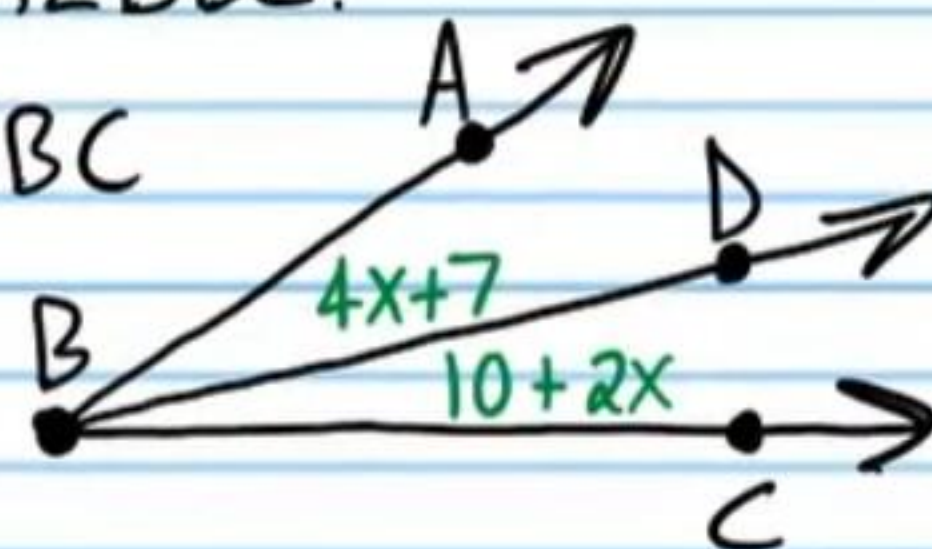
$$47 = 4x + 7 + 10 + 2x$$

$$47 = 6x + 17$$

$$30 = 6x$$

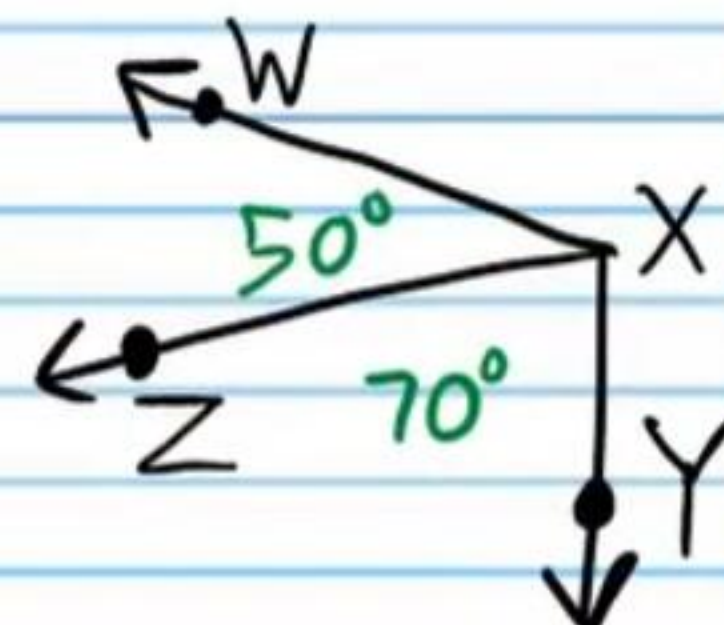
$$x = 5$$

ii) $m\angle DBC = 10 + 2(5)$
 $= \boxed{20^\circ}$



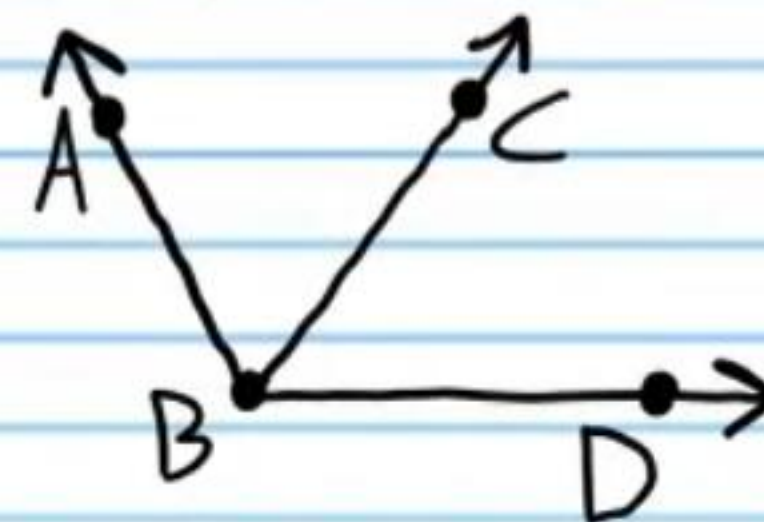
⑬ Find $m\angle WXY$.

$$\begin{aligned} m\angle WXY &= m\angle WXZ + m\angle ZXY \\ &= 50^\circ + 70^\circ \\ &= \boxed{120^\circ} \end{aligned}$$



⑭ If $m\angle ABC = 70^\circ$, and $m\angle CBD = 65^\circ$, find $m\angle ABD$.

$$\begin{aligned} m\angle ABD &= m\angle ABC + m\angle CBD \\ &= 70^\circ + 65^\circ \\ &= \boxed{135^\circ} \end{aligned}$$

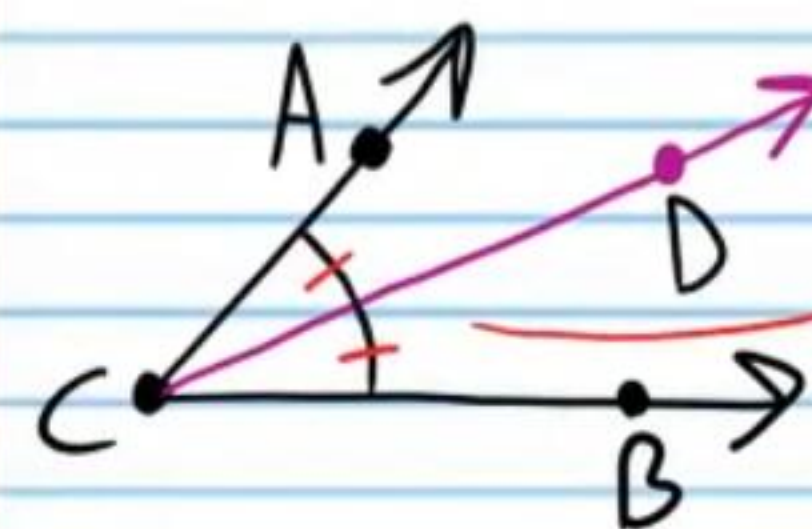


Congruence: Having the same size and shape (notated by $\cong$).

If two angles are congruent, then their measures are equal.

Example: If $\angle ABC \cong \angle DEF$, then $m\angle ABC = m\angle DEF$.

Angle Bisector: A ray that divides an angle in half, such that the two resulting angles are congruent.



These red marks indicate the angles $\angle ACD$ and $\angle DCB$ are congruent.

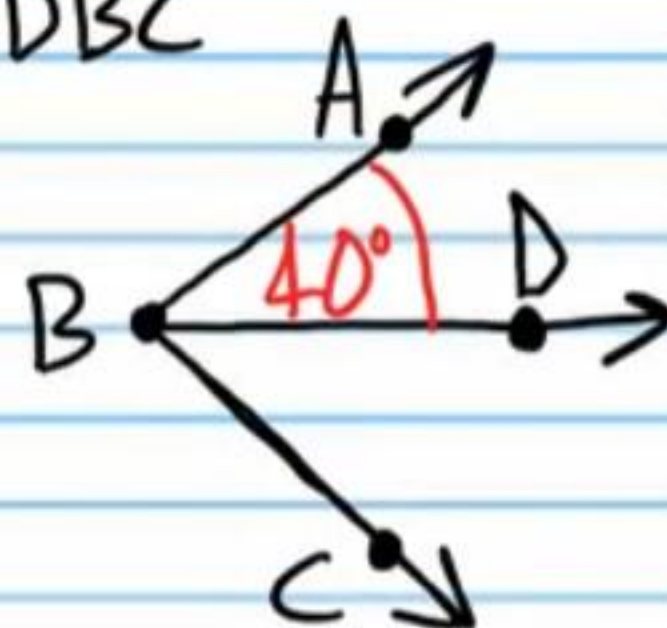
Practice

⑮ If $\vec{BD}$ bisects $\angle ABC$, find $m\angle DBC$

$$\angle ABD \cong \angle DBC$$

$$m\angle ABD = m\angle DBC$$

$$\boxed{40^\circ = m\angle DBC}$$



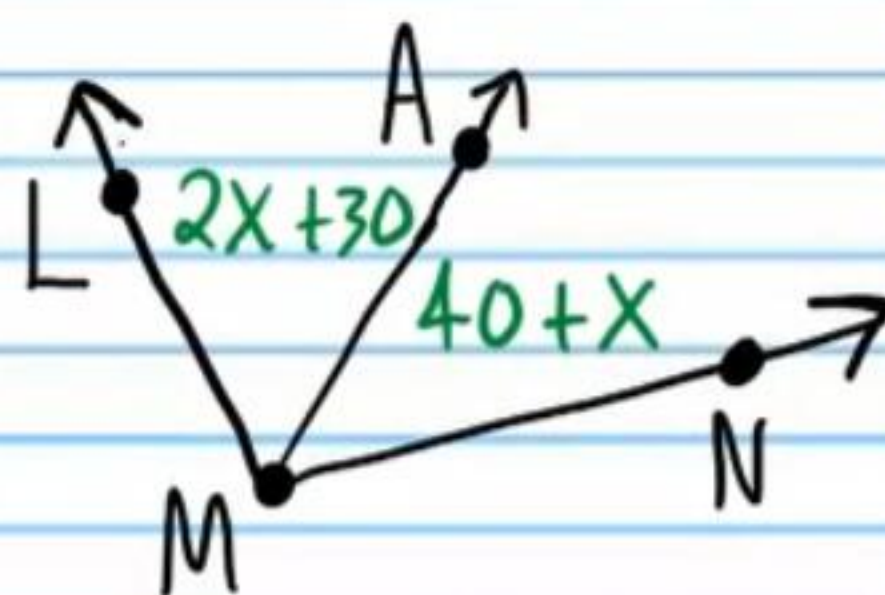
⑯ If $\vec{MA}$ bisects $\angle LMN$, find $m\angle LMN$.

$$i) \angle LMA \cong \angle AMN$$

$$m\angle LMA = m\angle AMN$$

$$2x + 30 = 40 + x$$

$$x = 10$$



$$ii) m\angle LMN = m\angle LMA + m\angle AMN = 2x + 30 + 40 + x$$

$$= 3x + 70$$

$$iii) m\angle LMN = 3(10) + 70 = \boxed{100^\circ}$$

⑰ If $\vec{BD}$ bisects $\angle ABC$, find $m\angle ABC$.

$$i) \angle ABD \cong \angle DBC$$

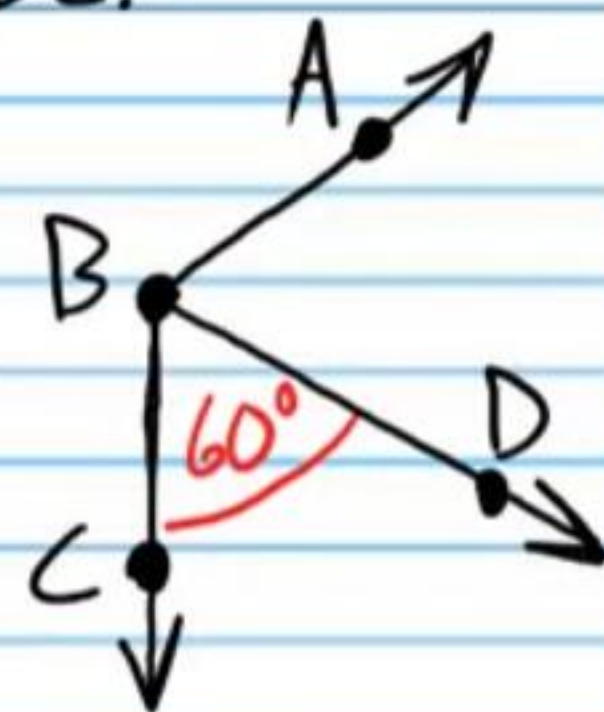
$$m\angle ABD = m\angle DBC$$

$$m\angle ABD = 60^\circ$$

$$ii) m\angle ABC = m\angle ABD + m\angle DBC$$

$$= 60^\circ + 60^\circ$$

$$= \boxed{120^\circ}$$



⑱ If $\vec{MA}$ bisects $\angle LMN$, find $m\angle AMN$.

$$i) \angle LMA \cong \angle AMN$$

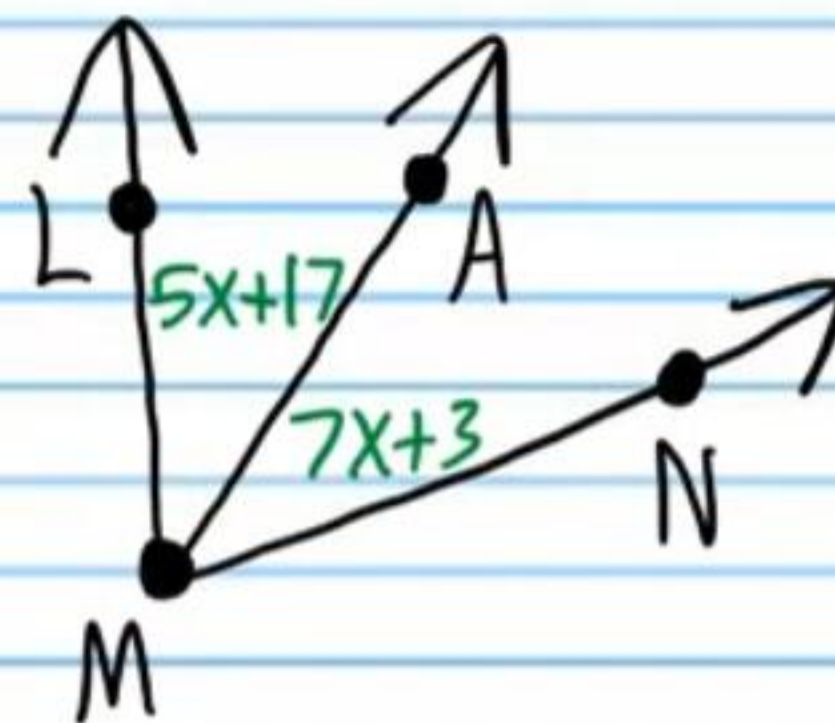
$$m\angle LMA = m\angle AMN$$

$$5x + 17 = 7x + 3$$

$$14 = 2x$$

$$7 = x$$

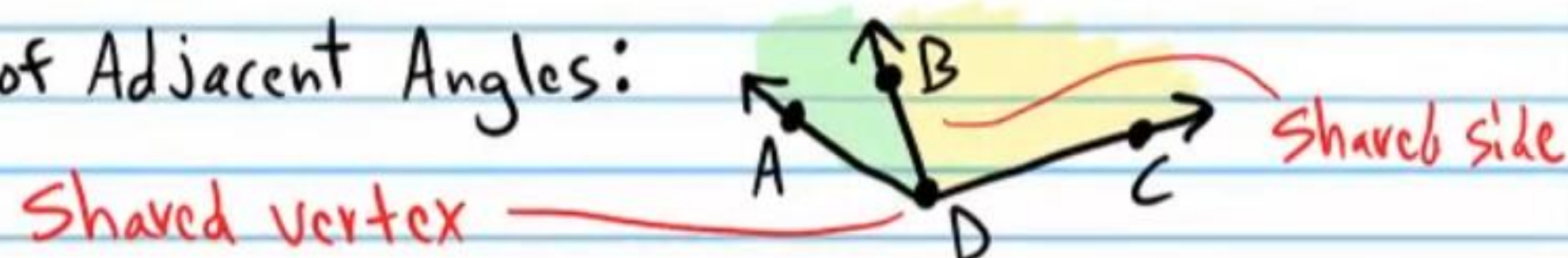
$$ii) m\angle AMN = 7(7) + 3 = \boxed{52^\circ}$$



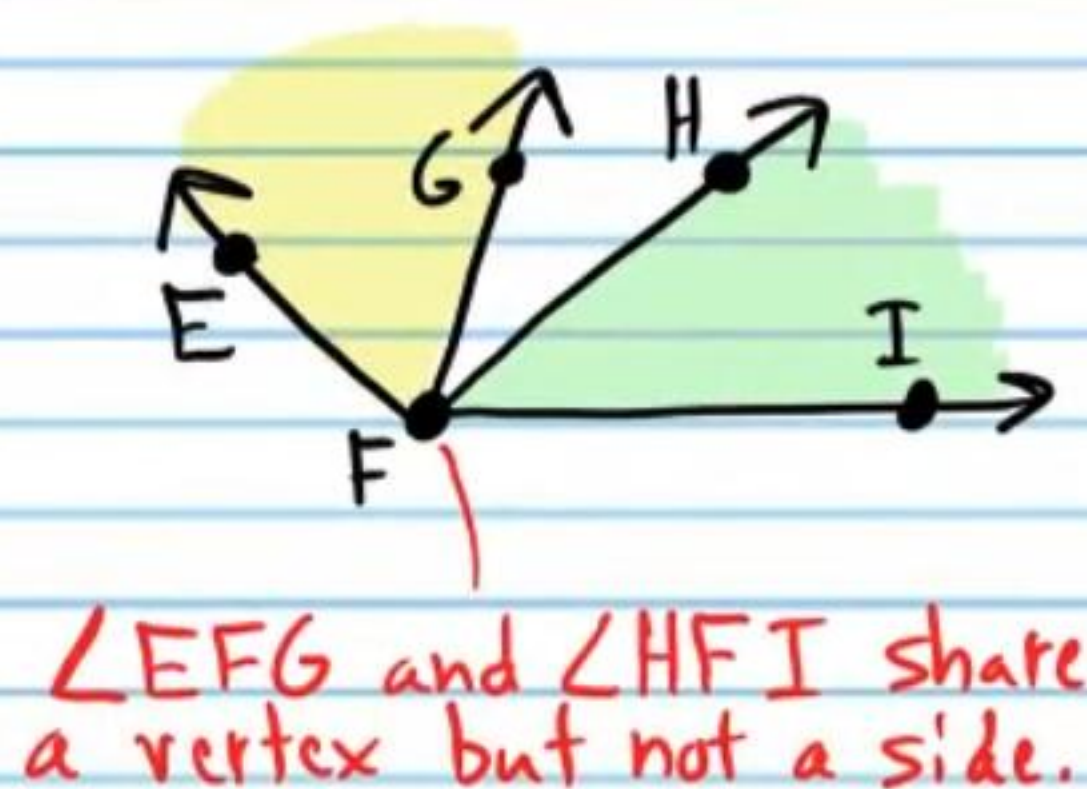
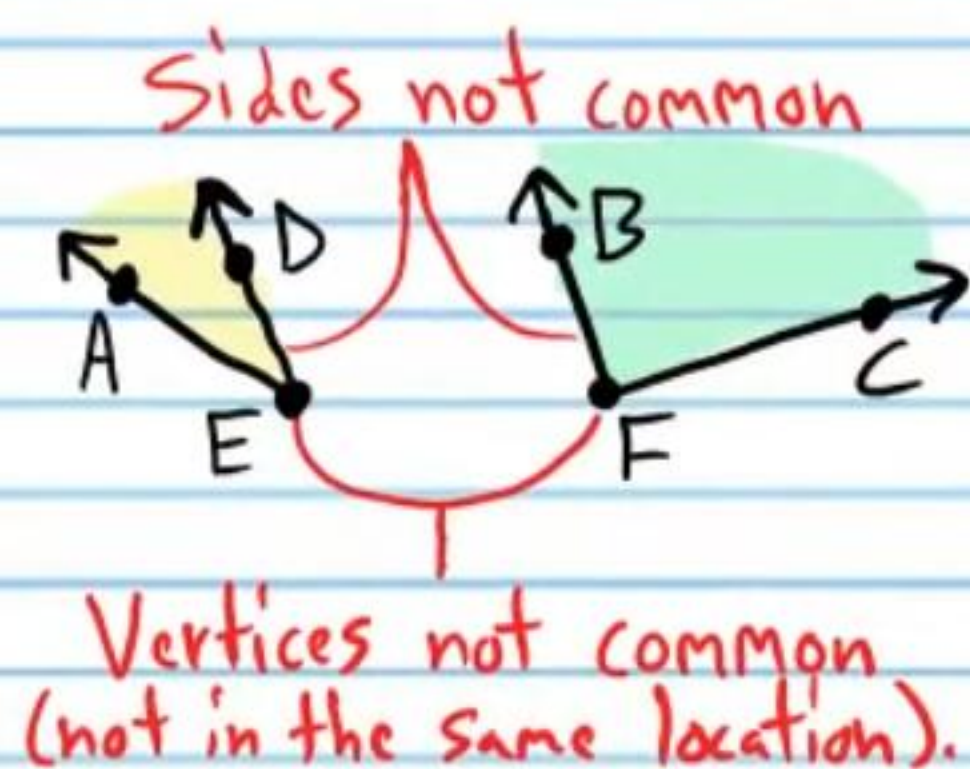
Angle Pairs

Adjacent Angles: Two angles that share a vertex and a side but have no common interior points.

Example of Adjacent Angles:



Examples of Nonadjacent Angles:



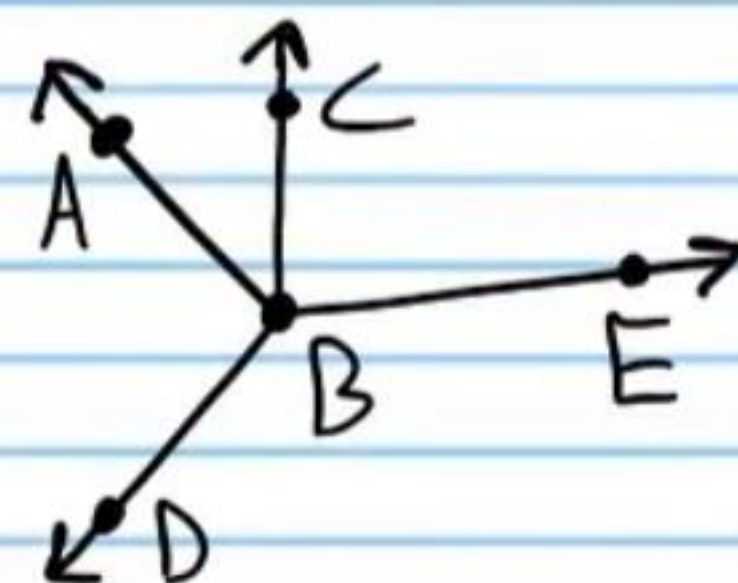
Practice

① a) Identify three pairs of angles in the illustration that are adjacent.

$\angle ABC$ & $\angle CBE$, $\angle ABC$ & $\angle ABD$, $\angle CBE$ & $\angle DBE$

b) Identify two pairs of angles in the illustration that are nonadjacent.

$\angle ABC$ & $\angle DBE$, $\angle ABD$ & $\angle CBE$

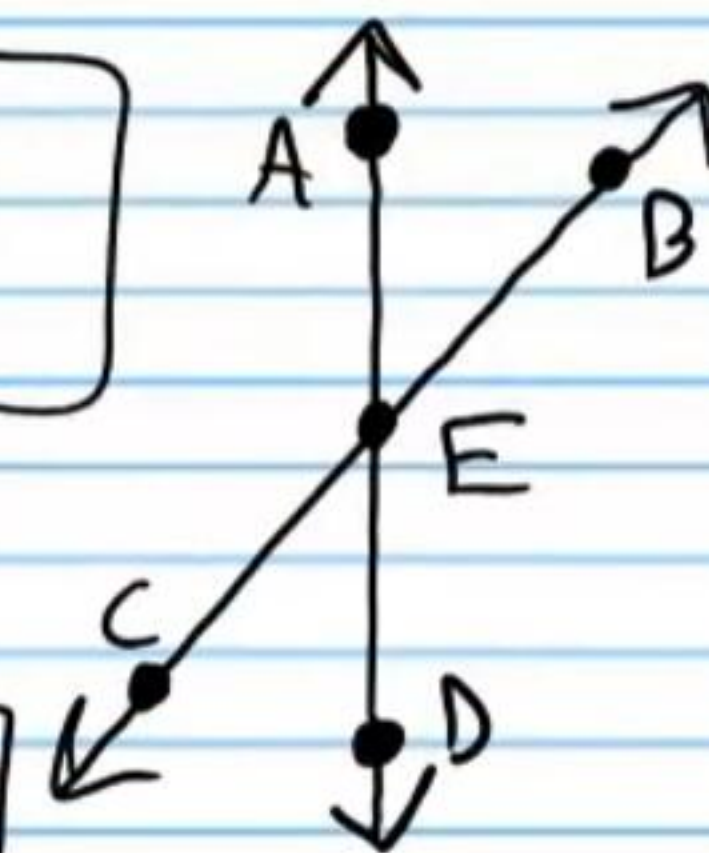


② a) Identify three pairs of angles in the illustration that are adjacent.

$\angle AEC$ & $\angle AEB$, $\angle AEB$ & $\angle BED$,
 $\angle CED$ & $\angle AEC$

b) Identify two pairs of angles in the illustration that are nonadjacent.

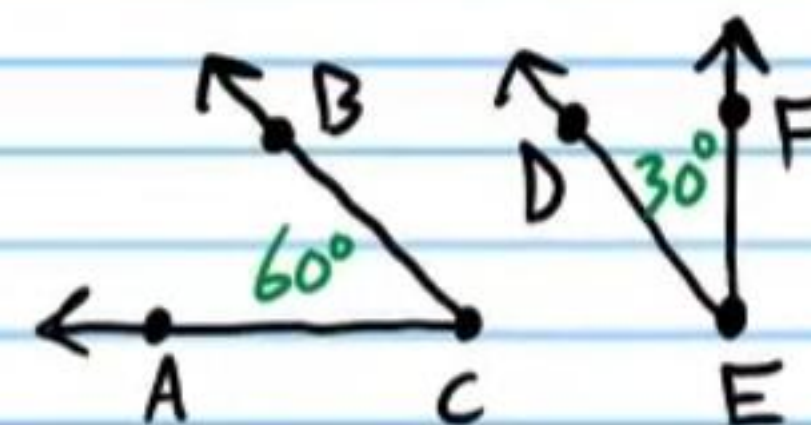
$\angle AEB$ & $\angle CED$, $\angle AEC$ & $\angle BED$



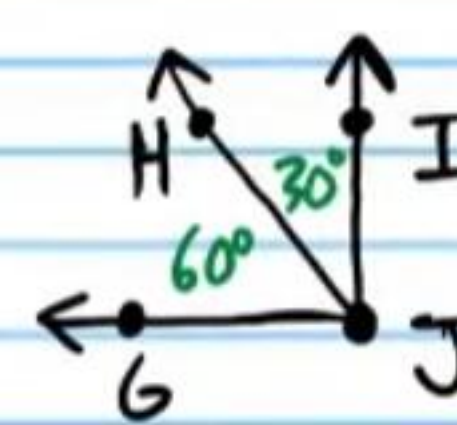
Complementary Angles: Two angles whose measures add up to 90° .

Examples of Complementary Angles

Nonadjacent



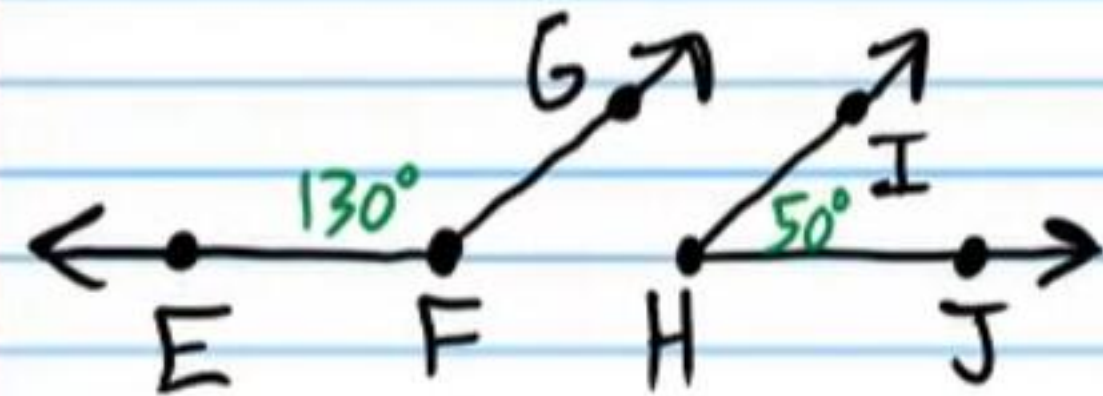
Adjacent



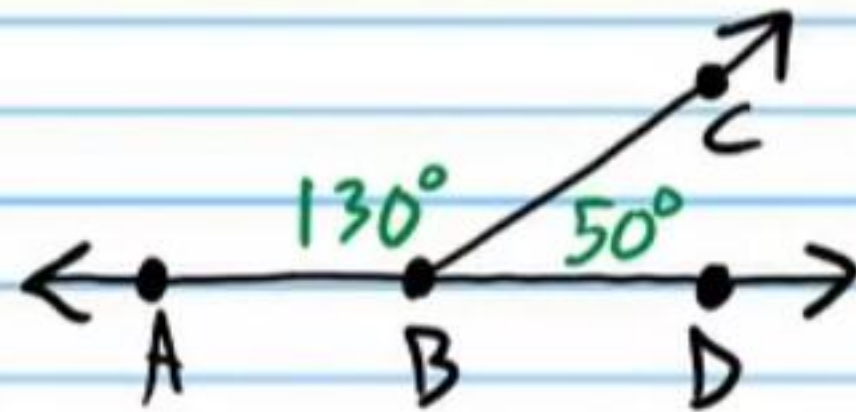
Supplementary Angles: Two angles whose measures add up to 180° .

Examples of Supplementary Angles

Nonadjacent



Adjacent



Practice

②① If two angles are complementary, and one of the angles is 75° , what is the measure of the other angle?

$$m\angle A + m\angle B = 90^\circ$$

$$75^\circ + m\angle B = 90^\circ$$

$$m\angle B = 15^\circ$$

$$\boxed{15^\circ}$$

②② $\angle A$ and $\angle B$ are supplementary. If $m\angle A = x - 5$ and $m\angle B = \frac{x}{2}$, find $m\angle B$.

$$i) m\angle A + m\angle B = 180$$

$$x - 5 + \frac{x}{2} = 180$$

$$\frac{3}{2}x - 5 = 180$$

$$\frac{3}{2}x = 185$$

$$x = 123.\bar{3}$$

$$ii) m\angle B = \frac{123.\bar{3}}{2} = \boxed{61.\bar{6}^\circ}$$

②③ If two angles are supplementary, and one of the angles is 60° , what is the measure of the other angle?

$$m\angle A + m\angle B = 180$$

$$60 + m\angle B = 180$$

$$m\angle B = 120^\circ$$

$$\boxed{120^\circ}$$

②④ $\angle A$ and $\angle B$ are complementary. If $m\angle A = 2x - 1$ and $m\angle B = x + 5$, find $m\angle A$.

$$i) m\angle A + m\angle B = 90$$

$$2x - 1 + x + 5 = 90$$

$$3x + 4 = 90$$

$$3x = 86$$

$$x = 28.\bar{6}$$

$$ii) m\angle A = 2(28.\bar{6}) - 1$$

$$= \boxed{56.\bar{3}^\circ}$$

②⑤ If the measure of an angle is 40° less than the measure of its supplement, what is the measure of the angle?

$$m\angle A + m\angle B = 180$$

$$m\angle A = m\angle B - 40$$

$$m\angle B - 40 + m\angle B = 180$$

$$2m\angle B - 40 = 180$$

$$2m\angle B = 220$$

$$m\angle B = 110^\circ$$

$$m\angle A = 110^\circ - 40^\circ$$

$$= 70^\circ$$

②6 If the measure of an angle is 10° more than 19 times the measure of its complement, what is the measure of the angle?

i) $m\angle A + m\angle B = 90$

$m\angle A = 10 + 19m\angle B$

ii) $m\angle A = 10 + 19(4^\circ)$

$m\angle A = \boxed{86^\circ}$

$10 + 19m\angle B + m\angle B = 90$

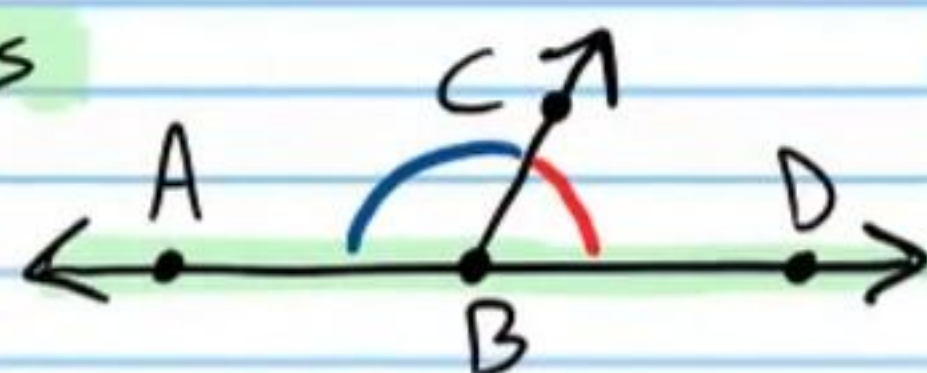
$10 + 20m\angle B = 90$

$20m\angle B = 80$

$m\angle B = 4^\circ$

Linear Pair: Two adjacent angles whose noncommon sides are opposite rays.

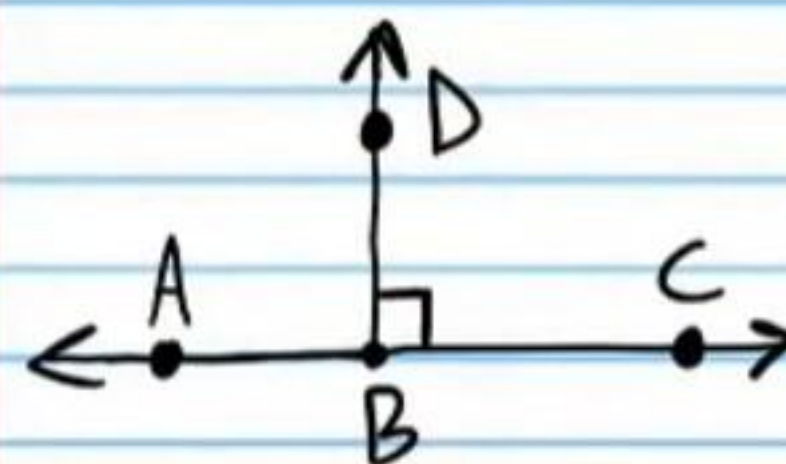
Opposite Rays



$\angle ABC$ and $\angle CBD$ form a linear pair.

Linear Pair Postulate: If two angles form a linear pair, then they are supplementary.

②7 Find $m\angle ABD$.

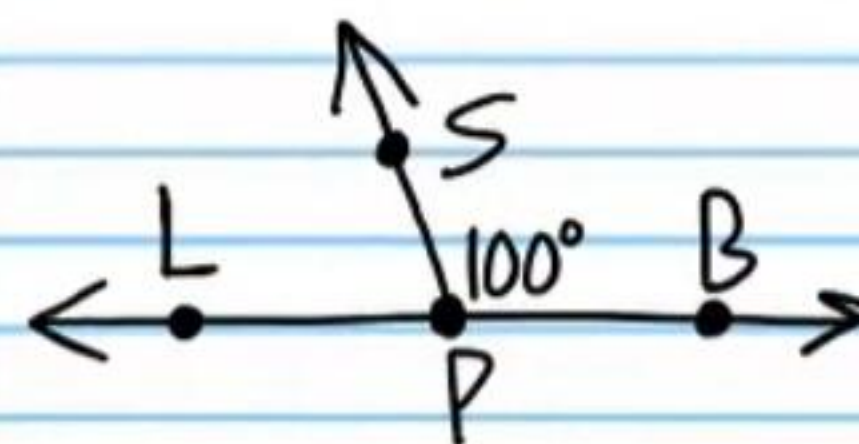


$m\angle ABD + m\angle DBC = 180$

$m\angle ABD + 90 = 180$

$m\angle ABD = \boxed{90^\circ}$

②9 Find $m\angle LPS$.

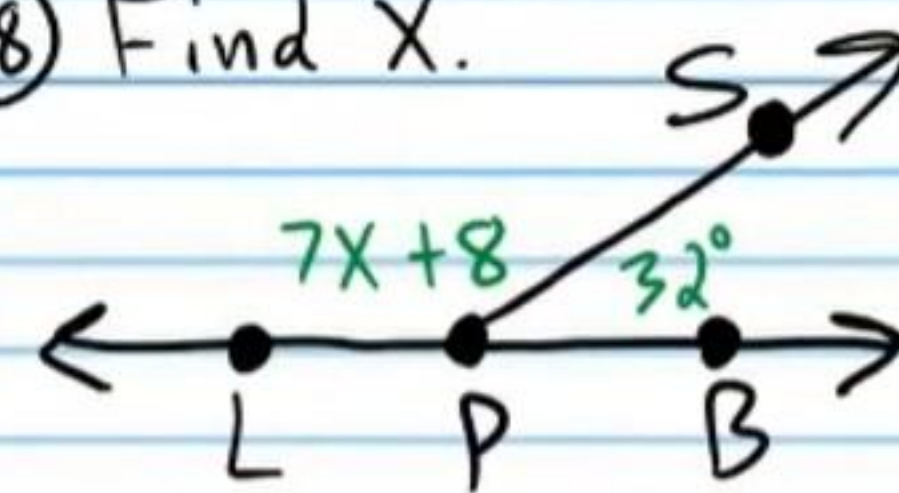


$m\angle LPS + m\angle SPB = 180$

$m\angle LPS + 100 = 180$

$m\angle LPS = \boxed{80^\circ}$

②8 Find X.



$m\angle LPS + m\angle SPB = 180$

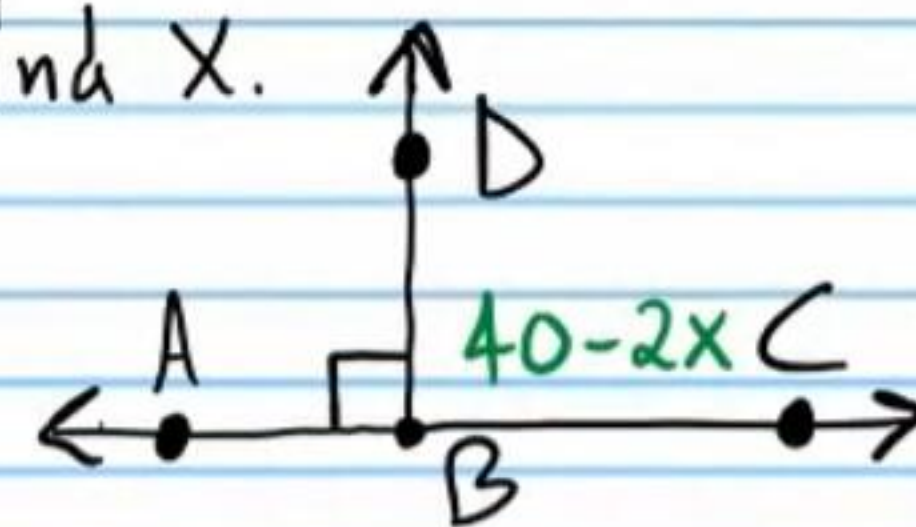
$7X + 8 + 32 = 180$

$7X + 40 = 180$

$7X = 140$

$X = \boxed{20}$

③0 Find X.



$m\angle ABD + m\angle DBC = 180$

$90 + 40 - 2X = 180$

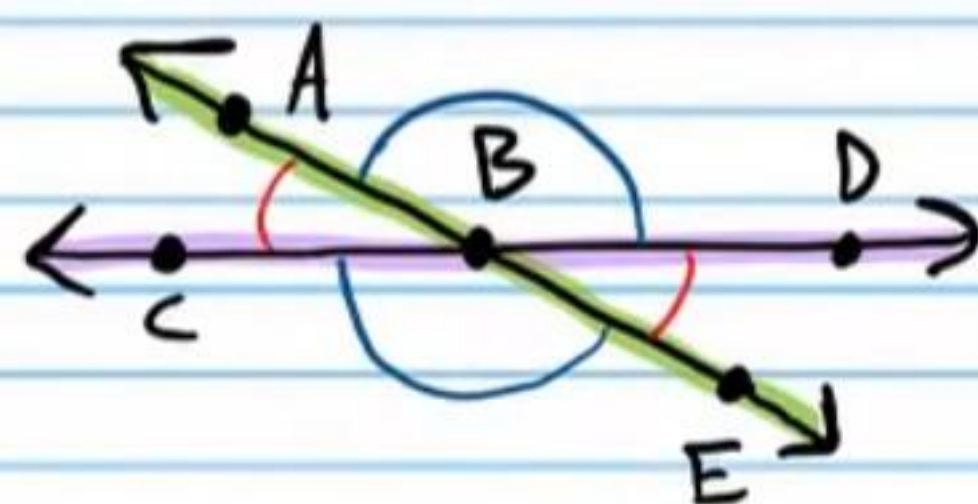
$130 - 2X = 180$

$-2X = 50$

$X = \boxed{-25}$

Vertical Angles: Two angles whose sides form two pairs of opposite rays.

Example:



$\angle ABC$ and $\angle DBE$
are vertical angles.

$\angle ABD$ and $\angle CBE$
are vertical angles.

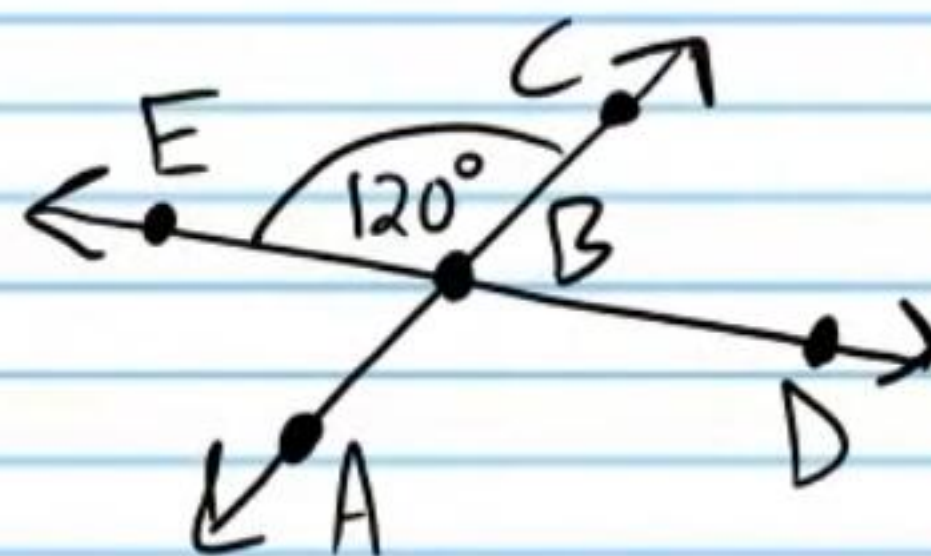
Vertical Angles Theorem: Vertical angles are congruent

③ Find $m\angle ABD$.

$$\angle EBC \cong \angle ABD$$

$$m\angle EBC = m\angle ABD$$

$$120^\circ = m\angle ABD$$



③ a) Find x . b) Find $m\angle DEB$.

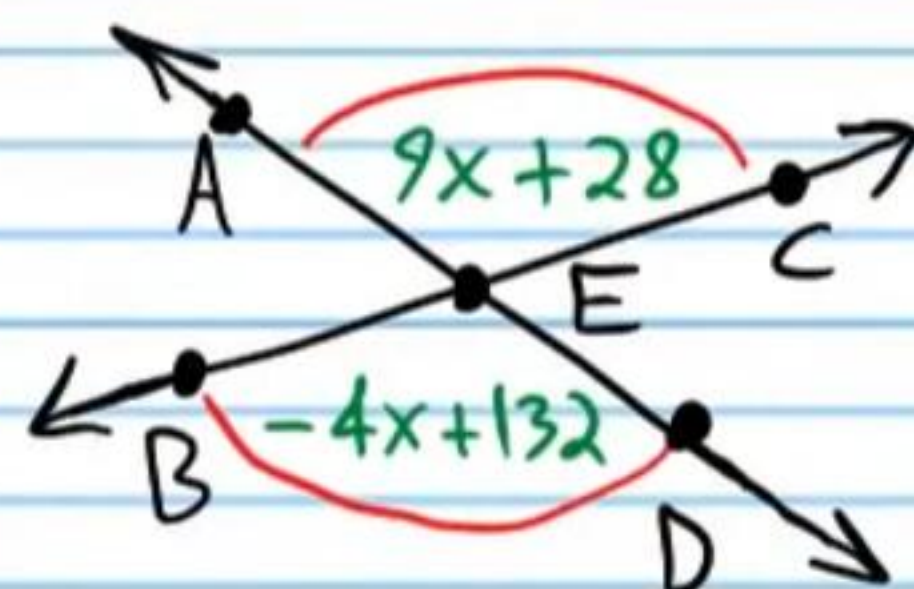
a) $\angle AEC \cong \angle DEB$

$$m\angle AEC = m\angle DEB$$

$$9x + 28 = -4x + 132$$

$$13x = 104$$

$$x = 8$$



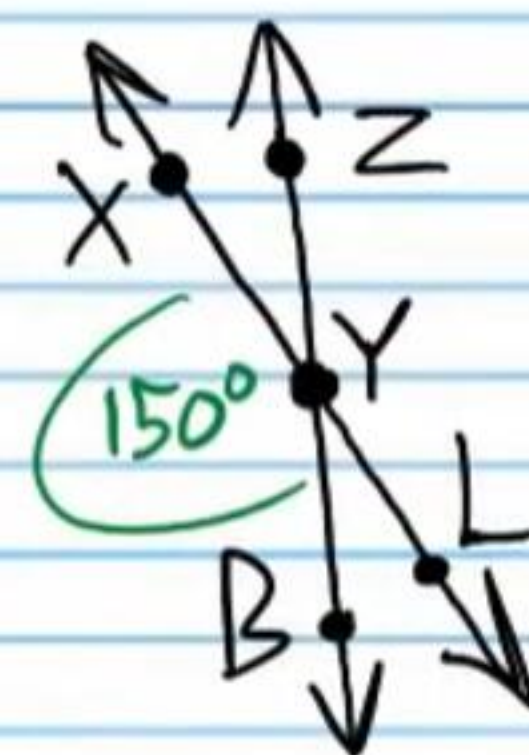
b) $m\angle DEB = -4(8) + 132$
 $= 100^\circ$

③ Find $m\angle ZYL$.

$$\angle ZYL \cong \angle XYB$$

$$m\angle ZYL = m\angle XYB$$

$$m\angle ZYL = 150^\circ$$



③ a) Find x .

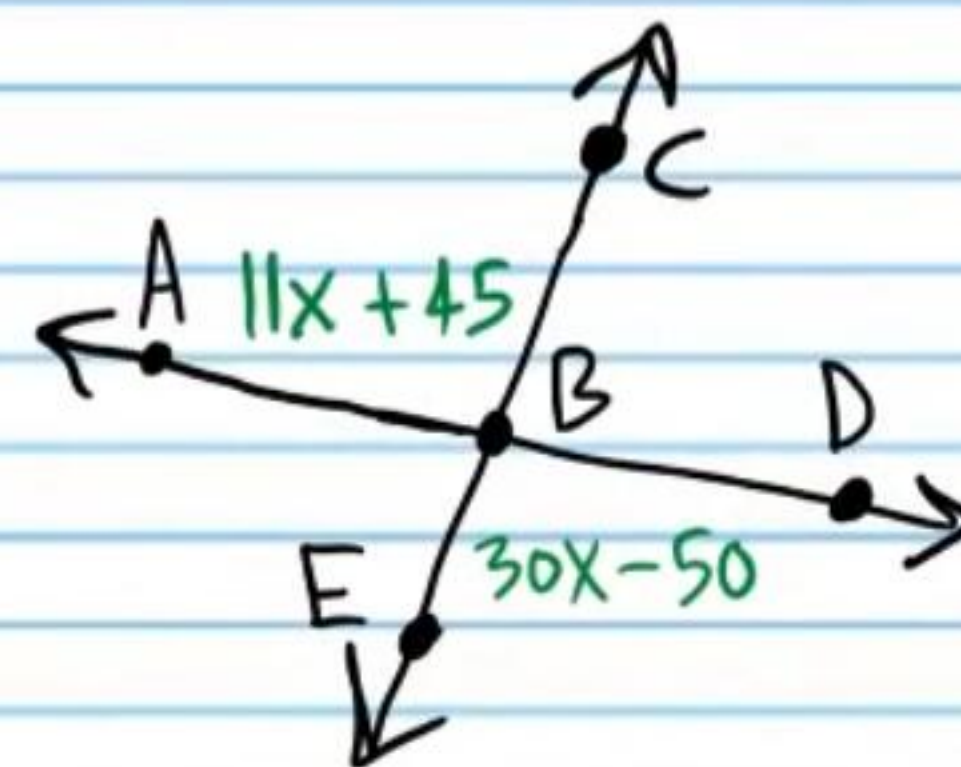
$$\angle ABC \cong \angle EBD$$

$$m\angle ABC = m\angle EBD$$

$$11x + 45 = 30x - 50$$

$$95 = 19x$$

$$x = 5$$



b) Find $m\angle EBD$.

$$m\angle EBD = 30(5) - 50 = 100^\circ$$